

EXECUTIVE SUMMARY

For decades, setting up a mountain bike's suspension has been guided by sag targets and feel. A rider bounces around the parking lot, consults a pressure chart, turns a few dials, and decides whether it feels right. It is an opinion rather than an informed decision. Engineers at FOX obsess about delivering the highest performing products but lack a rigorous way of knowing if the end customer is getting that performance.

Bill Brown posed the question: We deliver a gold-level product to the customer, but how do we make sure every FOX customer is riding the product to its highest potential? This project set out to answer what an ideal set up is, and why ideal can vary so much from rider to rider? We answer the question with simulation and data, and it has produced a result we believe is new to the industry.

We built two things. The first is a software model that simulates how any FOX fork and shock will behave on any bike, for any rider — how firm it is, how quickly it settles, and how well the front and rear are matched to each other. The second is a measurement system built around a GoPro camera and pose-tracking AI that determines, frame by frame, what a rider's weight distribution (center of gravity) is while they ride.

Putting the two together produced the central discovery of the project. **An experienced rider is continuously balancing the response of the bike with their body.** As a trail steepens, the rider shifts their weight rearward, keeping a familiar amount of weight on the front and rear tire contact patch, in effect, keeping the front and rear system responding at the same rhythm. We have now recorded multiple test riders confirming this hypothesis. Take two riders descending the same trail on different bikes, set up differently, and riding in visibly different positions. Both riders moved their bodies to different positions, but for their unique set ups it produced the same balanced response. As the hill steepens putting more weight on the front tire, an experienced rider will perfectly compensate by moving back and keeping a consistent weight on the front and rear. Likewise, the trail gets flat, rider moves forward, keeping that same confidence inspiring amount of weight on the front and rear wheel contact patches. We have duplicated this experiment multiple times, across multiple set ups and bikes, and confident enough now to say the goal of an ideal set up is a balanced natural frequency front and rear.

The takeaway is that the rider is the missing variable. The setup and the rider's body act as a mutual system, and a good setup is simply one the rider can keep in balance across the trails they actually ride. That ideal feeling from rider to rider is balance, and balance is something we can define, measure, and offer to riders with different riding position habits.

As far as we are aware, this is the first time this rider behavior has been measured directly and tied to a working setup model. For FOX it opens up several things no chart or factory baseline can: a physics-based definition of an ideal setup that we can stand behind, a way to capture feel as metrics and reproduce it on any other bike, and a path to recommending genuinely personalized setups to customers from a few simple questions. The remainder of this document is the formal engineering paper behind that result, written for a technical audience; this summary is intended to convey what was accomplished and why it matters.

A Single-Degree-of-Freedom Model and Computer-Vision Methodology for Mountain Bicycle Suspension Set-Up

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ABSTRACT

Mountain bicycle suspension set-up traditionally relies on manufacturer baselines and rider feel, with limited quantitative grounding for matching damper settings, spring rates, and air pressures to a specific rider, bike, and trail. This paper presents an analytical framework that addresses this gap and tests a central hypothesis: that an ideal set-up is one for which the rider's habitual riding position produces matching front-wheel and rear-wheel natural frequencies. As gradient disturbs the weight distribution, the rider corrects their center-of-gravity location to again balance the front and rear natural frequencies, a behavior we observe consistently in experienced riders. We develop two single-degree-of-freedom (SDOF) models for the front and rear suspension, parameterized by an air-spring formulation derived from the adiabatic gas law, a critical-damping percentage that normalizes feel across bike kinematics, and a friction model fit to laboratory and bounce-test data. To resolve the mass over each wheel during real riding, we develop a computer-vision system that uses a single GoPro camera and a pose-estimation pipeline to extract five body-segment metrics from the rider's torso. A random-forest regressor (machine learning technique) predicts rear-wheel bias (RWB) and rider-bike system center-of-gravity height (CGy) from these metrics, achieving coefficients of determination of 0.9 and better against a ground-truth pair of front and rear wheel scales. A multi-rider trail study across three trails of substantially different gradient (4.5%, 12.5%, and 42%) shows that riders on different bikes with deliberately different set-ups each move their center of gravity to a location that keeps the front and rear natural frequencies matched. We conclude that the rider acts as a feedback controller, and that the matching of front and rear natural frequencies — rather than any absolute frequency target — is the ideal that riders converge on.

Keywords: mountain bicycle, suspension, single-degree-of-freedom model, damping, leverage ratio, computer vision, center of gravity, random forest regression

NOMENCLATURE

A_{pos}, A_{neg}	positive and negative air-spring areas (m^2)
c	damping coefficient ($N \cdot s/m$)
c_{cr}	critical damping coefficient ($N \cdot s/m$)
$c_{cr}\%$	percentage of critical damping
CGy	rider-bike system center-of-gravity height (m)
F_{air}	air-spring force (N)
F_{bumper}	bottom-out bumper force (N)
F_{damper}	damper force (N)
FR	fork (front-wheel) subscript
RW	rear-wheel

FW	front-wheel
f_n	undamped natural frequency (Hz)
k	spring stiffness (N/m)
$L1$	vertical pixel length of torso (hip-midpoint to shoulder-midpoint)
$A1$	pixel area of torso polygon
$P1$	pixel perimeter of torso polygon
$S2S$	shoulder-to-shoulder pixel length
$S1$	vertical pixel length, shoulder midpoint to top of frame
LR	leverage ratio, dF_{shock} / dF_{wheel}
LR_{sag}	leverage ratio at sag
m	sprung mass over a single wheel (kg)
P_1, P_2	air-spring pressures before/after compression (Pa)
rFW, rRW	loaded radii of front and rear wheel (m)
RWB	rear-wheel bias, $RW / (RW+FW)$
V_1, V_2	air-spring volumes before/after compression (m^3)
VS_{ratio}	volume-spacer ratio, FW_{max} / FW_{sag}
ΔW_F	change in front-wheel weight with vehicle tilt (N)
γ	adiabatic coefficient (1.2 – 1.4 used as bounds)
θ	vehicle tilt angle (rad)

1. INTRODUCTION

Mountain bicycle performance is strongly influenced by the way the suspension is set up, and yet the task of selecting damper clicker positions, spring rates, air pressures, and volume-spacer counts for a given rider is treated, in industry practice, as a problem solved by chart lookups, manufacturer baselines, and rider intuition. The question that motivated this work was posed by VP of Engineering Bill Brown: what is an ideal mountain bicycle suspension set-up, how can we ensure our customers are riding with ideal settings for them, and, if they are not, how can we guide them there with confidence in our suggestions. The objective of the work described here is to define an analytical footing rather than leaving its answer to personal opinion.

We pursue the objective through two coupled efforts. The first is an SDOF mechanical model of the front and rear of the bicycle, executed inside a MATLAB App Designer tool we refer to as the Set-Up Application. The second is a computer-vision pipeline that estimates the rider's center-of-gravity location while riding actual trails, so that the SDOF mass term can be assigned an experimentally grounded value rather than a generic assumption. We are still learning and iterating, and the framework presented here is the current state of an ongoing internal research project.

The remainder of the paper is organized as follows. Section 2 develops the SDOF model and its four constituent force terms: spring, damper, mass, and friction. Section 3 describes the computer-vision system for estimating rear-wheel bias and system center-of-gravity height. Section 4 correlates the SDOF model against bounce-test data and then presents a multi-rider trail study covering three riders on three trails of substantially different gradient. Section 5 discusses the central question the framework was built to answer

— what is an ideal suspension set-up? — and frames the rider as a feedback controller that matches front-and-rear natural frequency through fore-aft and vertical body shift.

2. SINGLE-DEGREE-OF-FREEDOM MODEL

The rider mass is split into a front-wheel and a rear-wheel SDOF model, each of which obeys the equation of motion in Eq. (1). The mass m is a function of the trail gradient θ and the rider position on the bike. The damping force f_c is a function of velocity and position due to leverage ratio curve. The spring force f_k is a function of position. A separate friction force f_{friction} is added; for the fork this depends on both velocity and lateral load, while for the rear shock we treat it as a velocity-only term.

$$m(\theta) \ddot{x} + f_c(x, \dot{x}) + f_k(x) + f_{\text{friction}}(x, \dot{x}) = F_{\text{forcing}} \quad (1)$$

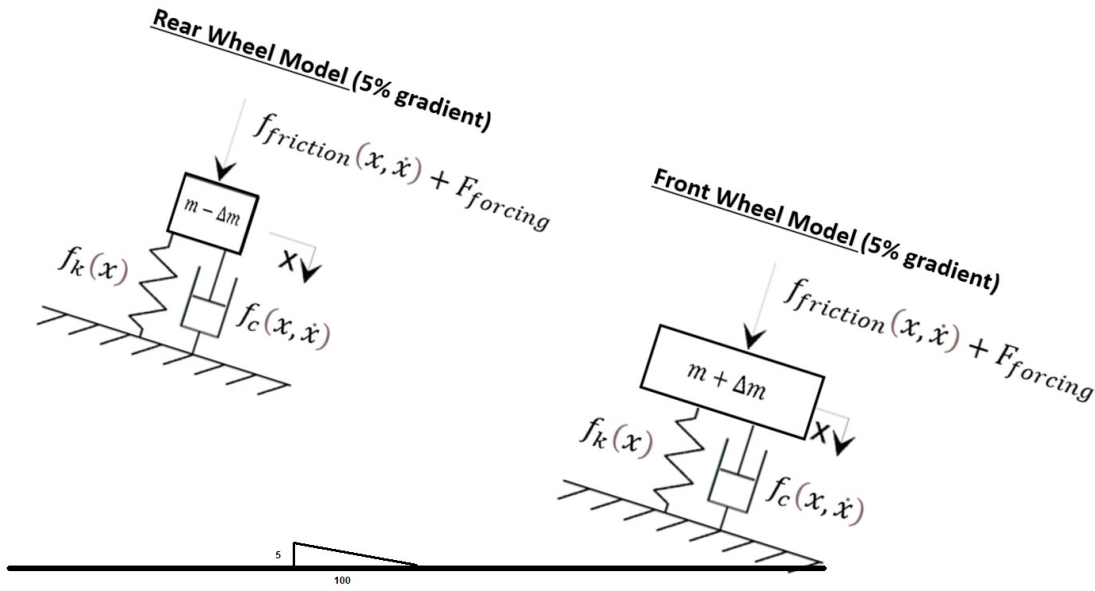


Fig. 1 Free-body diagram of the front-wheel and rear-wheel SDOF representations of the bicycle, including spring, damping, and friction terms (direction changes) and the gravity-driven weight shift that occurs at non-zero trail gradient.

2.1 Spring Force: Air Spring and Leverage Ratio

For air springs we use the adiabatic gas law. A discrete change in air-spring volume produces a new pressure given by Eq. (2). Positive and negative air-spring chambers are treated separately, and the bottom-out bumper force is added at the end of stroke. Eq. (3) gives the net spring force at the shaft. Each shock and each fork air spring in our internal database — over one hundred configurations — is given its own calibrated function; the methodology mirrors the air-spring spreadsheet calculators developed by FOX engineers.

$$P_2 = P_1 \cdot (V_1 / (V_1 - \Delta V_1))^\gamma \quad (2)$$

$$F_{\text{air}} = (P_{\text{pos}} \cdot A_{\text{pos}}) - (P_{\text{neg}} \cdot A_{\text{neg}}) + F_{\text{bumper}} \quad (3)$$

To map shaft forces to wheel-referenced quantities we use a per-bike dataset stored as *dataLVR*(shock stroke, wheel travel, leverage ratio, headtube angle). For each position of shaft stroke the rear-wheel

position and the instantaneous leverage ratio LR are known. The vertical wheel-rate force is then obtained from Eq. (4):

$$LR = F_{\text{shock}} / F_{\text{wheel}} = dx_{\text{wheel}} / dx_{\text{shock}} \quad (4)$$

Two compact derived quantities are useful for spring set-up communication: the bottom-out ratio $VS_{\text{ratio}} = FW_{\text{max}} / FW_{\text{sag}}$, which captures bottom out force versus sag force, and wheel sag targets. Both of these establish targets for fork and shock. Unique to the fork we use a metric defined as Wheel Rate Skew (WRskew), which establishes a unitless stiffness comparison between front and rear wheel. This is an instrumental relationship for front to rear balance.

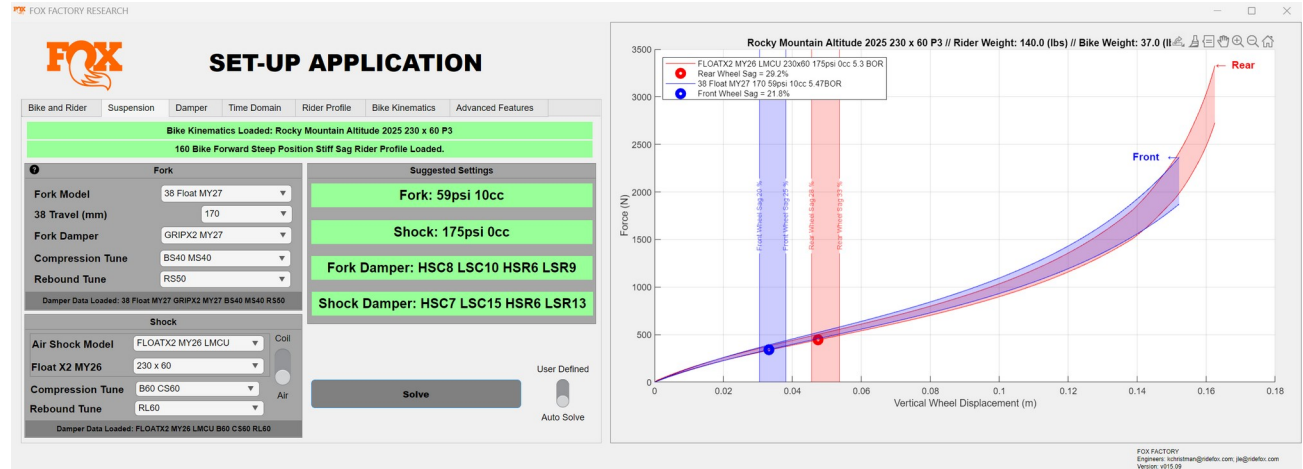


Fig. 2 Computed wheel-rate force versus vertical wheel displacement for a representative trail bike, with shaded bands corresponding to adiabatic coefficients of 1.2 (lower bound) and 1.4 (upper bound). Front (fork) and rear (shock plus bumper) curves are shown, and the calculated front and rear sag positions are marked.

2.2 Damping Force: From Settings and from Critical Damping

A damper force-versus-velocity model for the fork and shock are constructed from a swept dynamometer (PVP) dataset. To provide a dimensionally normalized handle on damper feel from the perspective of the wheel, we compute a critical-damping percentage. Critical damping for a single wheel is given by Eq. (5); the shaft-referenced critical damping in Eq. (6) accounts for the leverage ratio at sag. The percentage of critical damping at any velocity is then Eq. (7).

$$c_{\text{cr},\text{FW}} = 2 \cdot \sqrt{(k_{\text{FW}} \cdot m_{\text{FW}})} \quad (5)$$

$$c_{\text{cr},\text{damper}} = c_{\text{cr},\text{FW}} \cdot (LR_{\text{sag}})^2 \quad (6)$$

$$c_{\text{cr}\%} = F_{\text{damper}}(\dot{x}) / (c_{\text{cr},\text{damper}} \cdot \dot{x}) \quad (7)$$

Percent critical damping is a unitless parameter that lets us speak about damper feel across various riders and bikes. A 200-lb rider running 110 psi in a FOX 38 fork and a 100-lb rider running 70 psi can target the same critical-damping percentages and obtain a damper response proportional to their respective mass and spring rate. This formulation draws on quarter-car methodology, introduced to me by Jimmy Le, and was implemented inside the Set-Up Application.

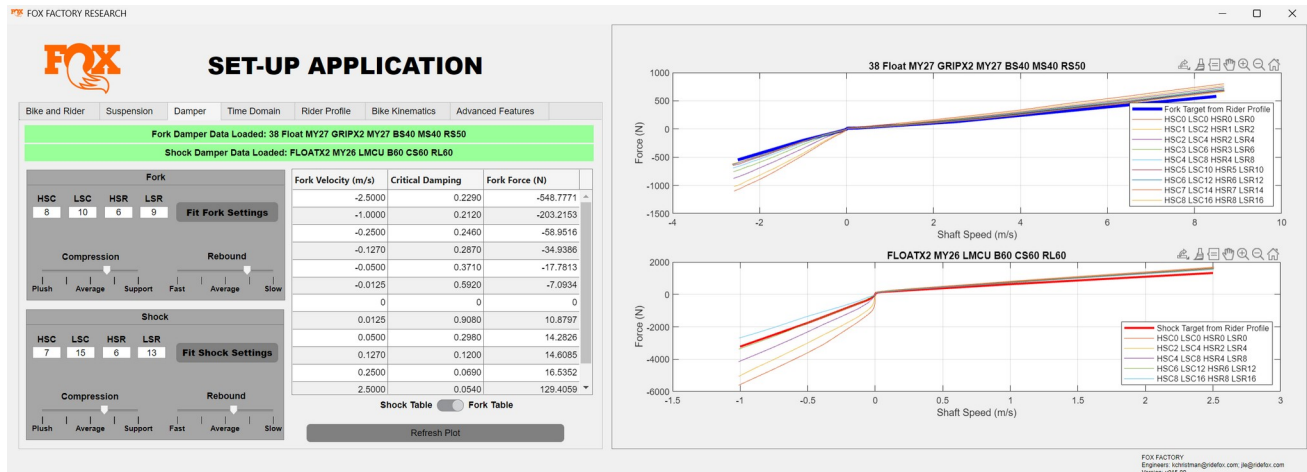


Fig. 3 Theoretical damper curve generated from a 4-way clicker input (red trace) overlaid on the swept dyno dataset. Force-versus-shaft-speed plots are shown for both the fork and the rear shock.

2.3 Friction

For the fork, sliding friction depends on both shaft speed and lateral (fore-aft) load. Lateral load at any position in travel is computed from the air-spring force, and a 2-D friction surface is then queried during the SDOF model solve. The lab-measured friction was developed by Andrew Taylor; the surface used in the model differs from the raw lab data in one important way: it excludes breakaway friction, because the model would otherwise interpret breakaway as a static force that exists at zero velocity and resists settling.

For the rear shock, friction is treated as a function of velocity only. We tested an extension that made rear friction depend on wheel force as well, but it did not improve correlation. Across all bounce tests, the rear-end friction term effectively behaves as a small additional damper.

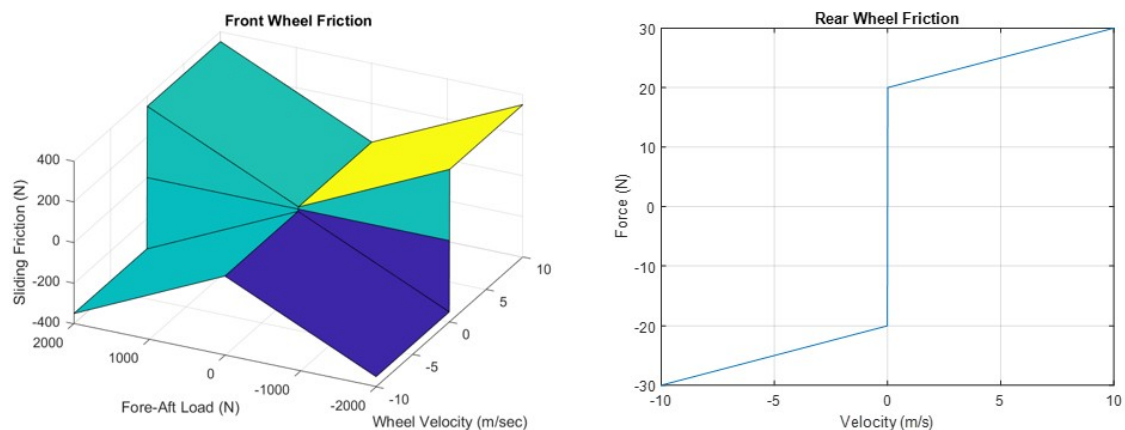


Fig. 4 Friction models used in the SDOF integration. Left: front-wheel sliding friction surface as a function of shaft speed and fore-aft load. Right: rear-wheel friction as a function of velocity only.

3. COMPUTER-VISION ESTIMATION OF RIDER POSITION

The rear-wheel bias and the system center-of-gravity height are the two scalars in the SDOF model that depend on the rider rather than on bike or component specifications. We measure both with a single GoPro camera mounted on the bicycle and a pose-detection landmark [2], calibrated against a pair of digital weight scales placed under the front and rear wheels. Igor Shvytov introduced the idea of using a vision system as a means of center of gravity approximation.

3.1 Instrumentation and Ground Truth

The instrumentation consists of a GoPro camera streaming at 30 Hz and two custom digital weight scales placed under the front and rear wheels, streaming at 10 Hz. The rear-wheel bias used as ground truth is computed directly from scale readings using Eq. (10). Using RWB rather than absolute weights reduces the influence of rider-induced vertical acceleration on the calibration data.

$$RWB = RW / (RW + FW) \quad (10)$$



Fig. 5 Instrumentation set-up. The bicycle is supported on two digital scales that report front and rear vertical loads at 10 Hz; a GoPro camera is fixed to the bike to capture the rider's upper body at 30 Hz.

3.2 Body-Segment Metrics

From each frame of camera data, the pose-detection landmarks returns a set of body-segment landmarks. Five metrics are derived from the torso landmarks and used as model inputs:

- $L1$ — the vertical pixel length from the hip midpoint to the shoulder midpoint
- $A1$ — the pixel area of the torso polygon
- $P1$ — the pixel perimeter of the same torso polygon
- $S2S$ — the shoulder-to-shoulder pixel length
- $S1$ — the vertical pixel length from the shoulder midpoint to the top of the frame

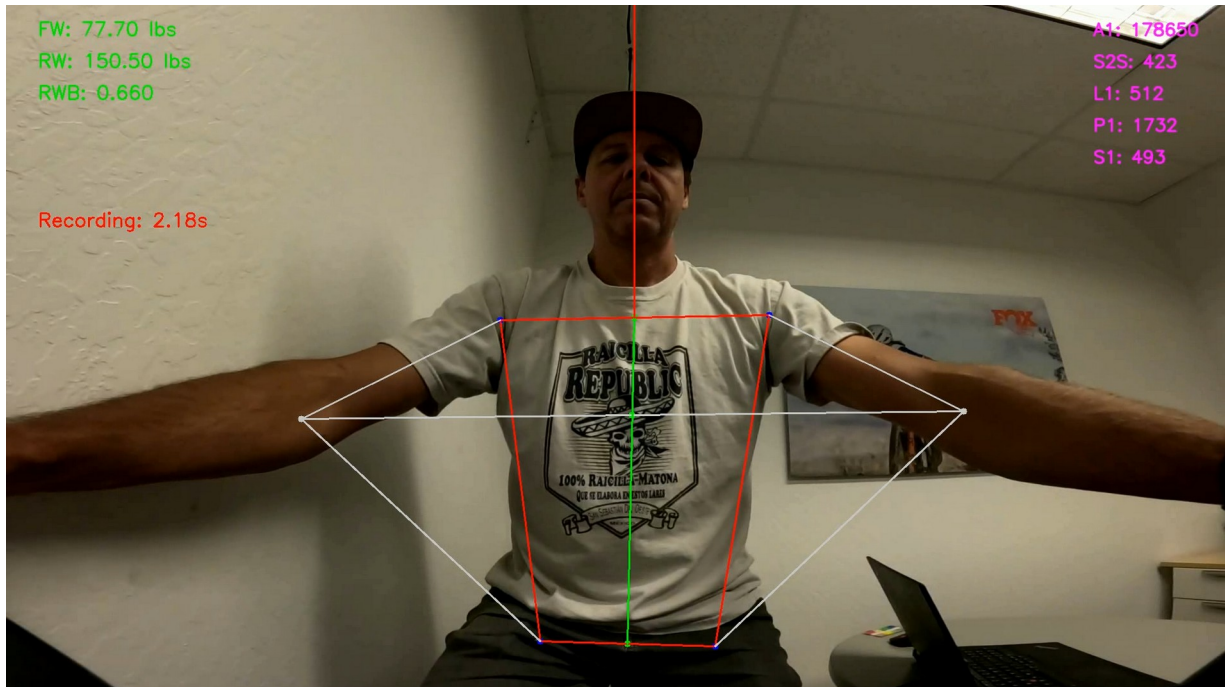


Fig. 6 Definition of camera-vision metrics on a single video frame. The red polygon is the torso region (perimeter $P1$, area $A1$); the green vertical segment is $L1$; the cyan horizontal segment is $S2S$; the dashed red vertical segment above the shoulders is $S1$.

3.3 Random-Forest Regressor for Rear-Wheel Bias

Calibration data are collected by having the rider move fore-aft on the bicycle on level ground while the camera and the scales record simultaneously. The rider repeats the fore-aft motion across several calibration rounds, producing multiple paired datasets of body-metric features and ground-truth RWB.

The RWB model uses seven input features: the five raw metrics ($P1$, $L1$, $A1$, $S2S$, $S1$) plus the second-order polynomial fits three of them ($A1_{pred}$, $P1_{pred}$, $L1_{pred}$). A random-forest regressor of 100 trees is trained with an 80/20 train-test split and 5-fold cross-validation; performance is reported in root-mean-square-error (RMSE) and R^2 . Feature-importance is then ranked. Often RWB is dominated by $P1$ and $A1$, which is intuitive: as the rider's torso projects more onto the camera plane, both perimeter and area scale with the change in fore-aft lean.

Across riders, R^2 ranges from 0.9 to 0.95. For one representative model, the standard deviation of the residual (actual RWB minus predicted RWB) is 0.012, so a prediction of $RWB = 0.70$ carries a 68% confidence band of ± 0.012 and a 95% band of ± 0.024 . This is precise enough that the model can be used downstream as a substitute for scale measurements during real riding. Further improvements have been made to filtering and model creation, below is data from the first week of model creation.

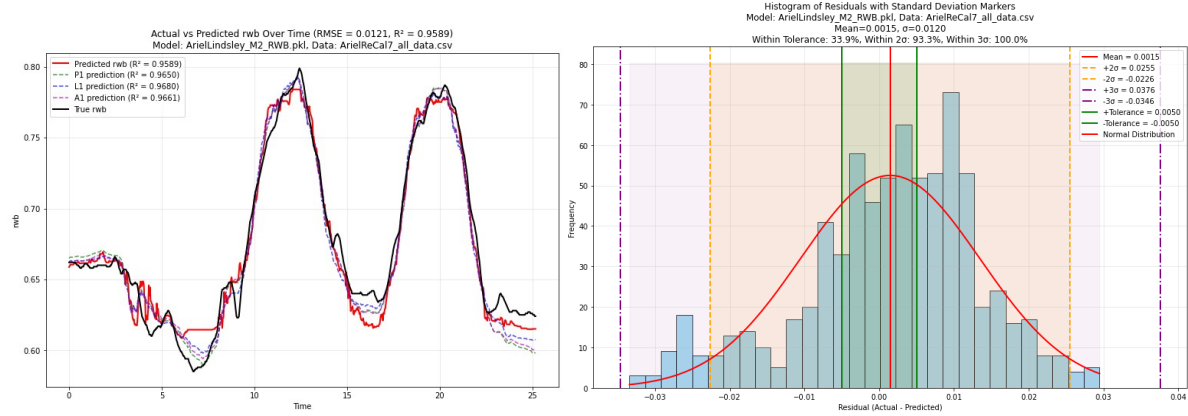


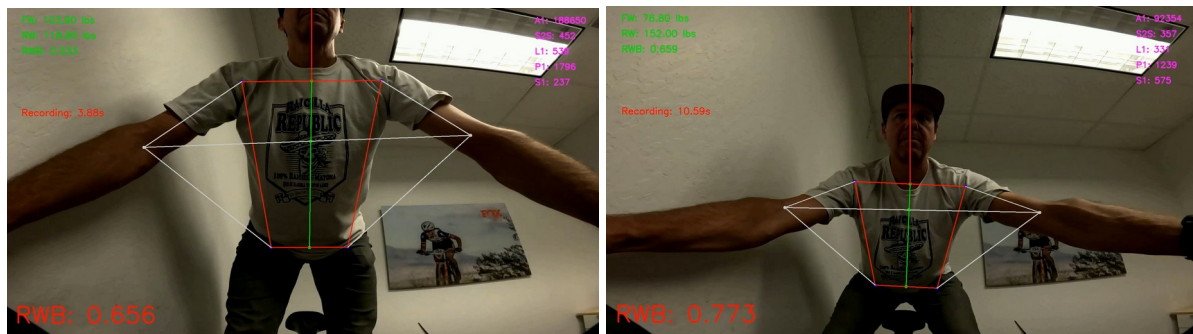
Fig. 7 Random-forest regressor performance on a held-out calibration round. Top: time-domain comparison of predicted RWB (red) versus measured RWB (black) over approximately 25 s of fore-aft movement, with the individual single-metric predictions overlaid as dashed lines. Bottom: histogram of residuals with normal-distribution overlay and $\pm 1\sigma$, $\pm 2\sigma$, $\pm 3\sigma$ markers.

3.4 Center-of-Gravity Height

Milliken's vehicle-dynamics text [1] gives the CG-height calculation for a rigid vehicle whose rear wheels are elevated relative to the front, accounting for unequal front and rear wheel radii. The rider is not a rigid body, so a literal application of the rigid-body formula will not work. We adapt the equation by treating the rider-bike system as effectively rigid only at the instant of measurement, using the relationship in Eq. (11). During calibration the rear wheel is elevated to a known angle θ , the rider re-runs the fore-aft sweep, and ΔW_F is read from the scales.

$$CG_y = (\Delta W_F \cdot \ell) / (W \cdot \tan \theta) + [rFW \cdot (1 - RWB) + rRW \cdot RWB] \quad (11)$$

During angled calibration the flat-ground RWB model is queried alongside the live scale data; the discrepancy between the flat-ground prediction and the angled measurement is what drives the CGy estimate. A second random-forest regressor is trained on the same feature set with CGy as the target. Feature importance for the CGy model is dominated by *S1* — the vertical pixel length above the shoulders — which is intuitive because *S1* is more sensitive to vertical translation of the rider. Two- σ residuals for the CGy model are approximately ± 100 mm.



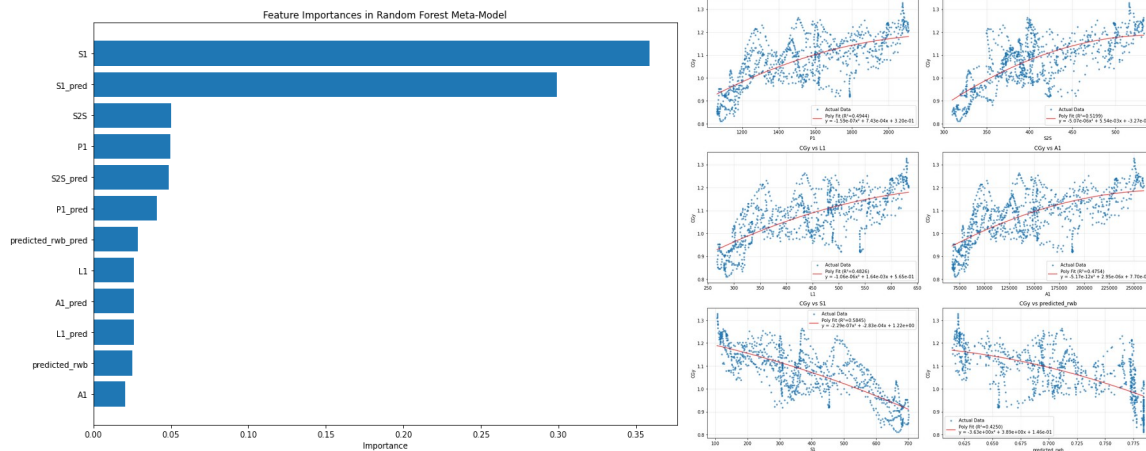


Fig. 8 CG-height estimation. Upper Left and Upper Right: RWB model overlayed on CGy model predicting CGy height. Bottom-Left: feature-importance ranking for the CGy random-forest model, with S1 and its polynomial fit dominating. Bottom-Right: CGy versus predicted RWB, showing the side-view geometric trace of the rider-bike center of gravity as the rider moves from seated to standing positions.

4. MODEL CORRELATION AND VALIDATION

4.1 Bounce-Test Validation of the SDOF Model

To validate the assembled SDOF model — spring, damping, mass, and friction — a Transition Patrol test bike was instrumented with linear potentiometers on both the fork and the rear shock. A rider applied a heavy, repeatable push on the bike and the system was allowed to settle. Bounce tests were run at three damper settings spanning the practical range: a heavy setting (HSC2 LSC4 HSR2 LSR4), a middle setting (HSC4 LSC8 HSR4 LSR8), and a light setting (HSC8 LSC16 HSR8 LSR16). For each test, the vertical wheel position calculated from the linear potentiometers was compared against the SDOF prediction with and without the friction term.

Two observations recur across the tests. First, including the friction term substantially improves decay correlation. At the lightest damper setting, the friction-free model predicts six or seven visible oscillations before the rear wheel settles, while the friction-augmented model settles in two or three, matching the recorded data. Second, the fork benefits more from the lateral-load-dependent friction surface than the rear shock benefits from its velocity-only friction term, because the rear-shock damping force is large enough that friction is a small fraction of total energy dissipation.

One remaining discrepancy is the natural frequency: the model consistently predicts a faster front-end and rear-end natural frequency than the data show. The likely explanations are that the effective rider mass during the bounce test is heavier than what is being assigned (it is difficult, as a rider, to hold yourself as a steady mass).

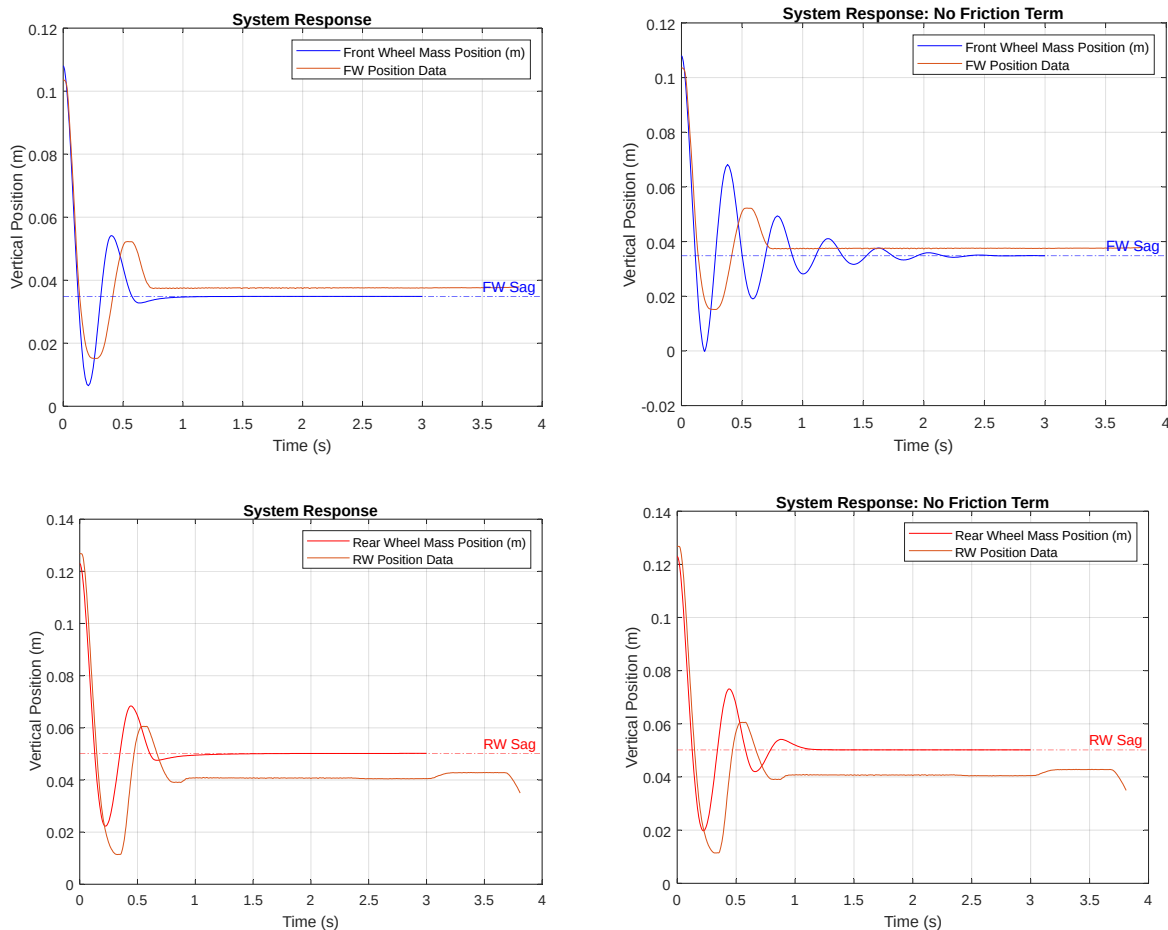


Fig. 9 Bounce-test correlation at HSC4 LSC8 HSR4 LSR8 (middle damper setting). Front-wheel and rear-wheel position are plotted versus time for the recorded data (orange) and the SDOF model with the friction term included (blue/red), and for comparison the same model with the friction term removed.

4.2 Multi-Rider Trail Study

Once the random-forest models for RWB and CGy and the SDOF model are individually constructed, the next step is to test on real trails with multiple riders. The following test has been run with multiple riders across various trails and gradients. We will focus on the three riders — Luke Hart, Ariel Lindsley, and Alex Petitemange — on two to three trails of deliberately different gradients: the Chup section at 12.5% average gradient, the Wiz end at 15%, and Wizard Chute at 42% average gradient. Each rider rode each trail multiple times, and a separate RWB and CGy random-forest model was constructed for each rider from a dedicated flat-ground calibration session. For each trail run, every video frame yielded a per-frame estimate of RWB and CGy, which we plot as a scatter in the RWB–CGy plane. Frames captured during jump take-off and landing produce a high-spread cloud that does not reflect a steady-state riding position; those frames are excluded from the balanced-position analysis.

All riders run different set-ups, when viewed with the set-up application metrics. Where Luke biases his fork soft and Ariel biases his fork stiff, Alex runs his front and rear wheel rates nearly line-on-line, with front and rear stiffness closely matched through travel. What most distinguishes Alex’s set-up from Luke’s

and Ariel's is sag: Alex runs noticeably deeper front and rear sag than either rider, while the front-to-rear relationship between his wheel rates is in between theirs.

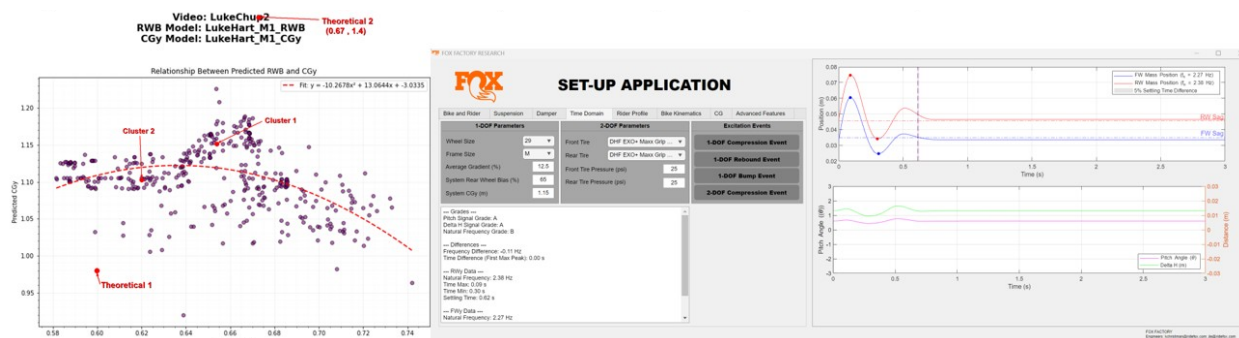


Fig. 10 Bike set up conveyed via wheel rates: (a) Ariel Lindsley, (b) Luke Hart, and (c) Alex Petittedemange.

4.2.1 Average Gradient Trail (12.5 %)

Luke's settled position cluster on Chup sits near $RWB = 0.65$ and $CG_y = 1.15$ m. Entered into the SDOF tool along with Luke's set-up, this position returns front and rear natural frequencies of 2.27 Hz and 2.38 Hz

— essentially balanced at the 0.1 Hz resolution of the solver. A neighboring sub-cluster at RWB = 0.62 and CGy = 1.10 m returns the same balanced response. Different combinations of RWB and CGy produce the same balanced natural frequency response, and Luke's habitual position lies within a balanced area rather than at a single point.



Ariel's settled cluster on the same trail sits at a substantially different point: roughly RWB = 0.62 and CGy = 1.17 m, taller and farther forward than Luke. Entered into the SDOF tool along with Ariel's set-up, this position returns front and rear natural frequencies of 2.38 Hz — the same number as Luke. The bikes are different, the set-ups are different, the riders' body positions are different, but each rider moves their center of gravity into a position that would yield a balanced front and rear natural frequency. Because Ariel's preference of a stiffer front wheel rate, we expect more mass is needed over the front wheel to balance that set up, which is the result we see.



Fig. 11 Same-trail, different-setup result on the Chup descent (12.5% gradient). Left: Luke's RWB–CGy scatter with his settled cluster marked, returning an essentially balanced 2.27/2.38 Hz response (at the 0.1 Hz resolution of the solver) from the SDOF tool. Right: Ariel's RWB–CGy scatter at a different cluster location, returning a balanced 2.38/2.38 Hz response from the same tool with his set-up entered.

Because Chup's jump features make a clean cluster difficult to isolate, a 15% segment referred to as Wizard End was recorded as a more consistent stand-in. On this segment Alex's positions cluster tightly around RWB = 0.62 and CGy = 1.3 m, and the SDOF tool again returns a balanced front-and-rear response at that position. To check the robustness of the CGy estimate, the same footage was evaluated with several rider models built from different calibration rounds; the predicted CGy varies modestly between models, but the balanced-position conclusion is unchanged.

On a 15% grade descent, Alex's footage was processed near RWB = 0.62 and CGy = 1.3 m. Entered into the SDOF tool with Alex's set-up, this position returns a balanced front-and-rear response, consistent with Luke and Ariel on a similar trail. As noted for the other riders, Chup carries the most trail noise of the

three segments — repeated drops and jump features inject body positions that do not represent steady-state riding — so a better trail location was used for Alex.

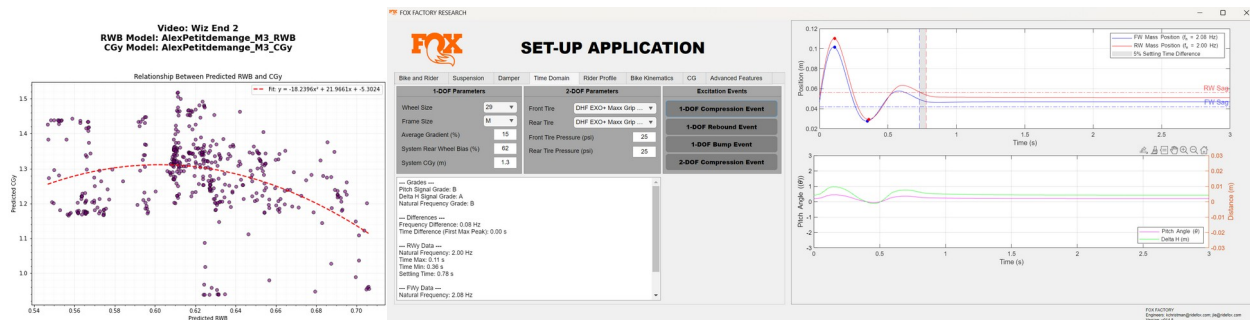


Fig. 12 Alex Petittedemange on the Wizard End segment (≈15% gradient), recorded as a cleaner stand-in for the jump-heavy Chup descent. Left: RWB–CGy scatter with the settled cluster near RWB = 0.62 and CGy = 1.3 m. Right: the SDOF tool returning a balanced front-and-rear response at that position with Alex's set-up entered.

4.2.2 Steep Gradient Trail (42%)

Wizard Chute, at a 42% gradient, presents a much narrower band of survivable positions; the riders' clusters are notably tighter than on Chup. Ariel's settled position shifts dramatically — well off the back of the bike, with a markedly different RWB and a lower CGy — and yet the SDOF tool, queried at this position with Ariel's set-up, again returns 2.38 Hz at the front and rear. Across a more than threefold change in trail gradient, the rider's body shift restores the same balanced response.



Fig. 13 Ariel Lindsley on Wizard Chute (42% gradient). Left: RWB–CGy scatter showing the settled position shifted well rearward and lower than on Chup. Right: the SDOF tool returning a balanced 2.38/2.38 Hz response at that position with Ariel's set-up entered.

Luke is further off the back of the bike and lower which again makes sense for his set. His time domain response result is 2.27 Hz at the front and 2.38 Hz at the rear, close to but not exactly matching, hovering at the 0.1 Hz resolution limit of the model.

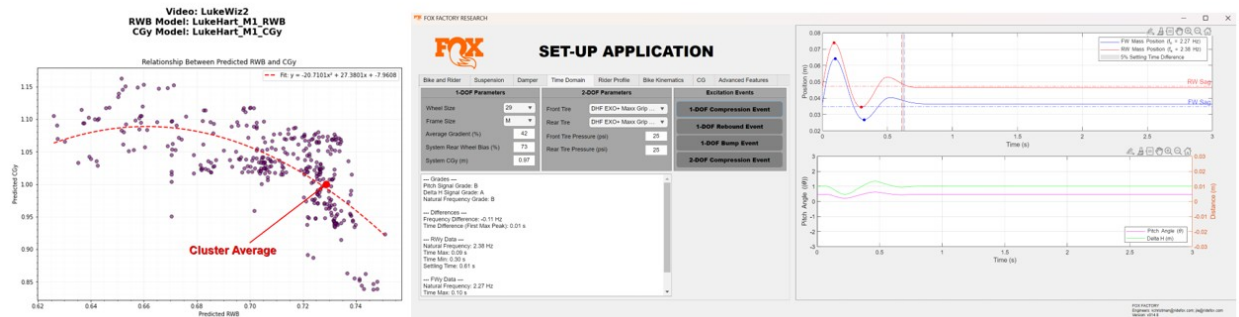


Fig. 14 Luke Hart on Wizard Chute (42% gradient). Left: RWB–CGy scatter with the settled cluster farther off the back and lower. Right: the SDOF tool returning 2.27 Hz front and 2.38 Hz rear — close to matched, at the 0.1 Hz resolution limit — with Luke’s set-up entered.

On Wizard Chute (42% gradient) Alex again settles into a balanced position, even though his body position there is markedly different from his Chup position; the steep gradient forces every rider into a narrow band of survivable stances. To probe the result further, the compression-event velocity in the Set-Up Application was increased to confirm that balance is maintained as more stroke is used, not only at the low-speed settling condition.

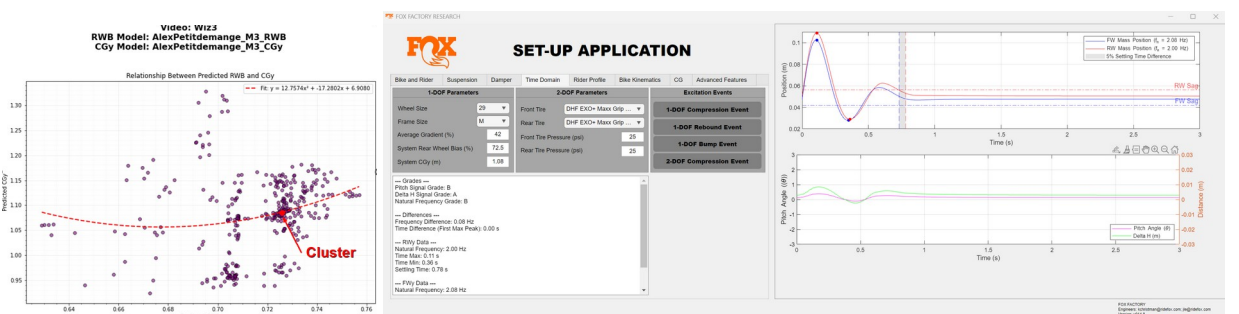


Fig. 15 Alex Petittedemange on Wizard Chute (42% gradient). Left: RWB–CGy scatter with the settled cluster marked. Right: the SDOF tool returning a balanced front-and-rear response with Alex’s set-up entered, confirmed as compression-event velocity is increased.

Across three gradients, and with a set-up that differs from Luke’s and Ariel’s in both sag depth and wheel-rate skew, Alex converges on positions that balance his front and rear natural frequencies, providing a third independent confirmation that the rider moves to restore front-and-rear balance regardless of the specific set-up.

5. DISCUSSION: WHAT IS AN IDEAL SUSPENSION SET-UP?

The framing question of this work was the one posed in the introduction: what is an ideal mountain-bicycle suspension set-up, and how can it be specified well enough to be reproducible for a given rider and bike? The framework developed in Sections 2 and 3, validated against bounce-test data in Section 4.1, and exercised across riders and trails in Section 4.2, supports a two-part answer.

5.1 The Simple Answer

A mountain-bike rider intentionally adjusts their riding position to maintain a familiar weight distribution on the front and rear contact patches as the trail gradient changes. The position of the rider,

combined with the wheel rates, determines the natural frequencies of the front-wheel and rear-wheel SDOF systems. When a rider returns to a position that feels right, they are also returning the two systems to the same natural frequency. An ideal set-up, on this account, is one for which the rider's habitual riding position produces matching front and rear natural frequencies over the range of trail gradients the rider actually rides.

5.2 *The Longer Answer*

The multi-rider study in Section 4.2 makes the simple answer more concrete and exposes its non-trivial structure. Three observations are worth restating.

First, the balance point is not a single position. Different combinations of RWB and CGy produce the same balanced natural-frequency response, so the rider's habitual position lies within a regime (area) in the RWB–CGy scatter plots rather than at a single point. The two clusters in Luke's Chup data are both balanced. This is consistent with a rider's actual experience of having a range of comfortable positions on the bike rather than a single locked-in stance.

Second, frequency is bike-and-set-up specific in its absolute value but riders-converged in its matching. The shock stroke and the spring rates set the absolute frequency band, and the rider then finds the position within that band at which the front and rear match. What the rider feels is therefore not a particular hertz value but the equality of front and rear response.

Third, a set-up has an envelope of gradients over which balance is reachable. Ariel's stiff-fork bias makes balance unreachable on a 4.5% fire road; no realistic forward position resolves the front-faster-than-rear mismatch. Luke's softer-fork, more rear-biased set-up extends his envelope down into the low-gradient range. Asking whether a set-up is good is therefore equivalent to asking whether its envelope of balanced gradients covers the trails the rider intends to ride.

5.3 *The Rider as a Feedback Controller*

The picture that emerges across the three observations is that the rider is acting as a feedback controller. The rider senses imbalance and shifts their mass to correct it. Over many hours of trail riding, this feedback loop produces a positioning habit that, in effect, is a tuned filter for matching the front and rear natural frequencies on the trails the rider commonly rides. The fact that two experienced riders converge on numerically identical balanced frequencies on a shared trail despite having different bikes, different set-ups, and different body positions is, in this view, exactly what one would expect from experienced riders.

Three practical handles follow from this picture. First, an average rider position is assumed RWB = 62 % and CGy = 1.1 meters, set ups can then be offered based on ideal gradient, which changes based on category of bike. Second, the rider-profile concept — a small set of normalized targets for sag, bottom out ratio, critical-damping percentage, and wheel rate skew — provides a starting set-up for a new rider whose preferences are easily determined via a few questions. Third, all rider profile metrics are unitless allowing an ideal set up to be transferrable to various bikes, rider weights, and disciplines.

6. CONCLUSION

This paper has developed an integrated framework for analytically grounded mountain bicycle suspension set-up and used it to test a specific hypothesis. The contributions are: (i) a single-degree-of-

freedom mechanical model of the front and rear suspension that combines an adiabatic air-spring formulation, a settings-interpolated damping model, a critical-damping-percentage normalization for wheel feel, and a friction model fit to laboratory and bounce-test data; (ii) a computer-vision pipeline that uses a single GoPro camera and a pose-estimation library to extract five torso-region metrics from each video frame, with a random-forest regressor predicting rear-wheel bias to within ± 0.024 (2σ) and system center-of-gravity height to within approximately ± 100 mm; and (iii) a multi-rider trail study covering three riders on three trails of substantially different gradient.

The central finding of the trail study is that experienced riders converge, through fore-aft body shift, on a position at which the front-wheel and rear-wheel natural frequencies match — regardless of the specific bike, the specific set-up, or the specific gradient. The rider is, in effect, the feedback loop that closes the suspension system, and a good set-up is one whose envelope of balanced gradients covers the trails the rider intends to ride.

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